DA351 – Explainable AI (XAI)

Course Project: Research Replication and Extension

Objective

The goal of this project is to introduce students to real-world XAI research by:

1. Understanding a published research paper
2. Reproducing its experiments
3. Identifying a research gap
4. Proposing and implementing an improvement
5. Writing a research-style paper

This project is designed to build research thinking, reproducibility skills, and scientific writing ability.

Team Formation

- Team size: Maximum 6 students
- Individual projects are not allowed
- Each team must clearly divide responsibilities and maintain contribution logs

Project Phases and Evaluation

Phase 0: Paper Selection

Marks: Mandatory approval (no marks)

Task

- Select one research paper in XAI

Paper Requirements

The paper must be from one of the following: - Q1 Journal OR - A / A* Conference OR - Faculty approved research paper . The paper should be published on or after 2023.

Deliverable

Submit a one-slide proposal including: - Paper title and link - Problem statement of the paper - Dataset used - XAI method used - Reason for selecting this paper

The paper must be approved by the faculty.

Deadline: Aug 28, 2026

Phase 1: Replication of Research Paper

Marks: 20 Marks (Based on the quality of the research paper, number of experiments done, similarity of your results with respect to the research paper results)

Task

Reproduce the key experiment of the selected paper.

Expected Work

- Implement the model used in the paper
- Implement the XAI method
- Use same dataset
- Reproduce at least two to three main result (table/graph)

Deliverables

- GitHub repository with code
- Result comparison with original paper
- 1-page summary of replication

Deadline: Sep 29, 2026

Phase 2: Research Gap Identification and Extension

Marks: 20 Marks (Based on the extension idea, supported with literature study, experiments and results with respect to the extension idea, research paper writing)

Task

Identify a limitation / extension in the original paper and propose a solution.

Expected Work

- Clearly define research gap
- Propose and implement one improvement (supported by good reference from literature)
- Show experimental comparison (before vs after improvement)

Deliverables

- Code updates
- Experimental results
- 1 page research gap explanation

Deadline: Nov 3, 2026

Academic Integrity and AI Usage Policy

AI Tools (ChatGPT, Copilot etc.)

Free to allow any AI tools. But ,

Each team must submit an AI_USAGE.md file containing: - Tool used - Prompts used - How output was modified

Plagiarism Policy

- Final research paper similarity must be less than 20%
- All sources must be cited
- Copying code without citation is not allowed

Violation will result in zero marks.

Reproducibility Requirement

Each project must include: - GitHub repository - Requirements file - Dataset link - Instructions to run code - Random seed mentioned

Top-performing projects will be:

- Guided for conference/workshop submission
- Supported by faculty for publication